

## Lab Class NLP:III

To solve the following task, please make yourself familiar with Cohen's Kappa, Fleiss' Kappa, and Krippendorff's Alpha first.

## Exercise 1 : Validity and Inter-Annotator Agreement

Fill in the following cloze. Use the terms from the box below. Terms may be used multiple times and not all terms must be used. Refer to [NLP:II-168 ff.] for definitions.

validity • reliability • stability • replicability • accuracy • consistency • measurements • agreement  
• annotation scheme • repeated measurements • identical conditions • sound principles

\_\_\_\_\_ refers to the quality of being well-founded on fact and established on \_\_\_\_\_.

\_\_\_\_\_ refers to the degree to which \_\_\_\_\_ of the same subject under \_\_\_\_\_ yield \_\_\_\_\_ results. For example, two annotators applying the same guidelines to the same texts should produce similar annotations.

Annotation \_\_\_\_\_ implies \_\_\_\_\_, the other way around does not hold: Two observers of the same event may well share the same prejudice while still being objectively wrong.

## Exercise 2 : Inter-Annotator Agreement

Assume you have collected sentiment assessments for a set of documents. Each document was judged by two annotators A and B as either “positive” or “negative”. The table below shows the agreement between the two annotators' judgements. Calculate Cohen's  $\kappa$  between the annotators.

|   |          | A        |          |
|---|----------|----------|----------|
|   |          | positive | negative |
| B | positive | 3        | 2        |
|   | negative | 1        | 14       |